

Lab Report Writing Guidelines

Each student group must submit a single lab report for every experiment conducted. All members of the group must contribute to the preparation of the report. The completed lab report must be submitted in the **immediate next lab class** following the experiment.

Report Submission Instructions:

- ✓ The report should be **handwritten**.
- ✓ Use **A4-size paper** for writing the report.
- ✓ Write on **both sides of each page** to conserve paper.
- ✓ **Maintain proper margins** on all pages.
- ✓ Use **graph paper** whenever graphs are required.
- ✓ **Label** all circuits, graphs and data points clearly and accurately.
- ✓ All submissions must be handed in at the **beginning of the class**.
- ✓ **LATE SUBMISSION IS NOT ALLOWED.**
- ✓ The **cover page** provided by the lab instructor must be used by every group. While all other parts of the report must be handwritten, the **cover page must be printed** — no handwritten information is allowed on it.
- ✓ Complete **all fields** on the cover page accurately. The **serial numbers** of students will be provided by the instructor.
- ✓ Use a **pencil and scale** to draw circuit diagrams or graphs neatly. Clear, well-labeled, and precise diagrams are required to obtain full marks.
- ✓ To achieve full marks, ensure that the report includes **all sections listed below**, presented in the **specified sequence**.
- ✓ Make sure what you write **makes sense** and falls in the realm of common sense.

Required Sections of the Lab Report:

1. **Cover Page** - Every lab report must include the prescribed cover page. The same format should be used for all reports. A sample cover page will be provided by the instructor.
2. **Objectives** - State briefly the **purpose or aim** of the experiment — in other words, what you intended to achieve or verify by performing it.
3. **Apparatus List** - Provide a simple, itemized list of all **apparatuses, instruments, and components** used during the experiment.
4. **Circuit Diagrams** - Draw clean and legible **circuit diagrams** using pencil. All components and connections should be **clearly labeled**.
5. **Data/Readings** - Present the **experimental data** collected during the lab session in a well-organized **tabular format**. Ensure that the table is neatly drawn and accurately represents the observed values.
6. **Questions and Answers** - Answer all **assigned report questions** clearly and concisely.
7. **Graphs** - Draw the necessary **graphs** using the recorded data. Label both axes, mark the data points clearly, and provide a concise **explanation** of the graph.

8. **Result Analysis** - Discuss the **results obtained** and state whether they align with the **objectives** of the experiment. Provide logical explanations for any discrepancies observed.
9. **Discussion** - This is one of the most important sections of the lab report. Your discussion should reflect your **understanding, observation, and analytical thinking** during the experiment. **Copying from others' discussions or previous reports will result in a zero mark.**
- **Focus on the following points:**
 - ✓ What did you **learn** from the experiment?
 - ✓ Were the **experimental results** consistent with theoretical expectations, or were there noticeable differences?
 - ✓ What **problems or difficulties** were encountered during the experiment?
10. **Attachment** - Finally, attach the **data sheet signed by the instructor** to your lab report.
11. **Multisim Simulation** - Add your **MULTISIM/LOGISIM** simulation output plots/results here.

Table: Lab Report Marks Distribution

Section	Marks
Cover Page	1
Margin every page	1
Objectives	1
Apparatus	1
Circuit Diagram	1
Data Table	2
Questions & Answers	5
Discussion	3
Attachment (signed data sheet)	1
Simulation	4